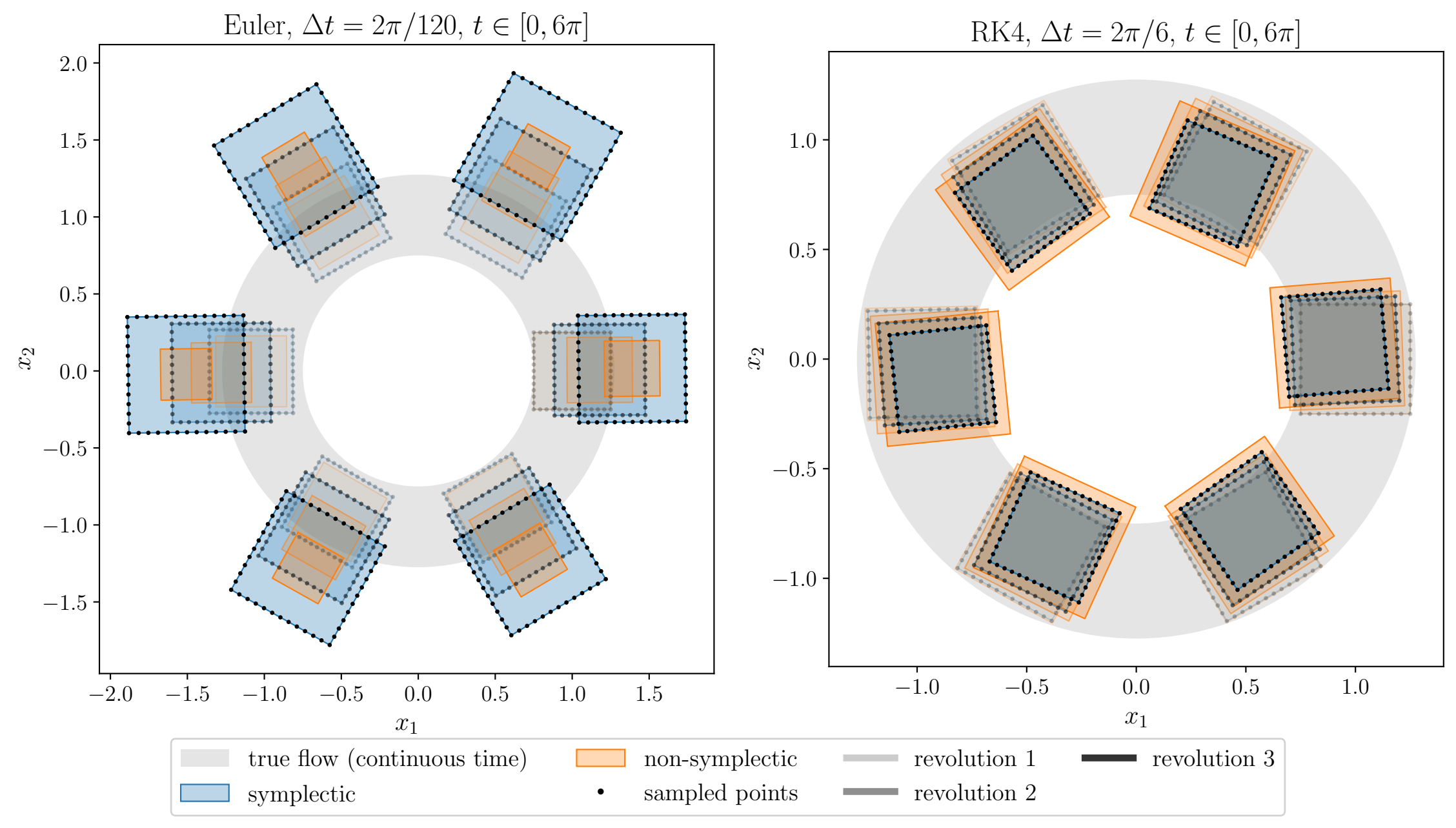

Euler, $\Delta t = 2\pi/120$, $t \in [0, 6\pi]$
RK4, $\Delta t = 2\pi/6$, $t \in [0, 6\pi]$
x_1
x_2
true flow (continuous time)
symplectic
non-symplectic
sampled points
revolution 1
revolution 2
revolution 3